

VentureNest Backend Architecture & Roadmap

Backend Bridge Specification & Mentor-Student Interaction Tracking Panel

1. Module Overview & Purpose

The VentureNest Backend Bridge API serves as the central orchestration service connecting the Web Admin Dashboard, Student Mobile App, and Mentors Mobile App. This document outlines the roadmap for future backend enhancements and the Mentor-Student Interaction Tracking System.

2. Future Feature: Mentor-Student Interaction Tracking Panel API

To enable real-time monitoring of incubator engagement, the backend will introduce a dedicated tracking endpoint (/api/admin/mentor-student-interactions) allowing administrators to track all active student-mentor relationships.

3. Interaction Tracking Data Contract

Metric / Field	Description & Data Source	Type
total_interacting_students	Count of distinct students with at least 1 mentorship request or review	Integer
total_active_mentors	Count of distinct mentors actively engaged with incubatees	Integer
active_pairs_count	Total active Student-Mentor pairing relationships	Integer
interaction_records	Array of detailed student-mentor interaction instances	JSON Array

4. Detailed Interaction Item Specification

Student Name	Mentor Name	Project Name	Messag es	Last Date	Status
Alex Rivera	Dr. Sarah Chen	EcoPack AI	6	2026-09-11	Active
Elena Rostova	Dr. Sarah Chen	MedFlow Tech	2	2026-09-10	Pending Mentor
Marcus Vance	Prof. David Miller	SolarGrid	9	2026-09-09	Resolved

5. Implementation Roadmap Steps

- Phase 1: Implement database view aggregating mentorship_requests and mentor_reviews.
- Phase 2: Add cached Redis endpoint GET /api/admin/mentor-student-interactions with pagination.
- Phase 3: Add WebSocket notification bridge when student submits help request or mentor submits feedback.
- Phase 4: Implement automated weekly engagement summaries for incubator administrators.